Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	8.2727	8	3.7441	count
ring_length_r30_px	headline	371.636	348	149.347	px
ring_length_r60_px	reported	978.909	929	338.093	px
owned_length_px	ownership-dependent	2009.92	1584.51	1063.17	px
edge_count	ownership-dependent	90	68	56.4624	count
mean_edge_length_px	ownership-dependent	24.271	23.3016	4.1168	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	12006.2	12118	2958.11	px^2
ring_area_r60_px2	normalised	28554.3	29514	6304.69	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0300733	0.0290736	0.00746476	px^-1
ring_density_r60	normalised	0.0336962	0.0345758	0.00810109	px^-1
roots_per_100px_perimeter	normalised	2.4334	2.3689	0.6696	per 100 px
ring_attached_length_r30_px	network v2	286.727	312	153.719	px
ring_attached_length_r60_px	network v2	664.364	557	363.465	px
ring_length_w_r30_px	network v2	414.325	398.777	171.806	px
ring_length_w_r60_px	network v2	1105.47	1032.28	386.087	px
sholl_crossings_r10	network v2	9.0909	8	5.0091	count
sholl_crossings_r20	network v2	13	12	5.0794	count
sholl_crossings_r30	network v2	15	15	5.2536	count
ring_width_mean_r30_px	width	6.5261	6.5998	0.6532	px

Per field
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measure	value	units
analysed_area_px2	1.0074e+06	px^2
total_skeleton_length_px	34260	px
canalicular_length_density_per_px	0.0340082	px^-1
junction_count	701	count
junction_density_per_px2	0.00069585	px^-2
lacuna_count	12	count
lacunae_per_px2	1.191e-05	px^-2
skeleton_component_count	277	count

mean_component_length_px	123.682	px
median_component_length_px	27	px
field_length_density_w_per_px	0.0384608	px^-1
field_density_without_flagged_per_px	0.0338477	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3706	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	32773	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	194	525	1057.78	63	16.7901	6.0839
2	18	655	1394	2598.19	128	20.2983	7.0918
3	6	309	852	884.813	36	24.5781	6.9032
4	8	468	1205	3517.01	152	23.1382	6.3053
5	9	435	1102	3371.64	177	19.0488	7.4759
6	7	303	828	1241.47	50	24.8294	6.5001
7	8	513	1421	2815.84	141	19.9705	6.9486
8	8	311	929	1062.22	33	32.1886	6.6332
9	7	348	851	1584.51	68	23.3016	5.5018
10	11	420	1250	3061.9	135	22.6807	6.5998
11	5	223	635	1131.06	42	26.9301	6.5688
12	4	103	301	840.465	28	30.0166	5.2586

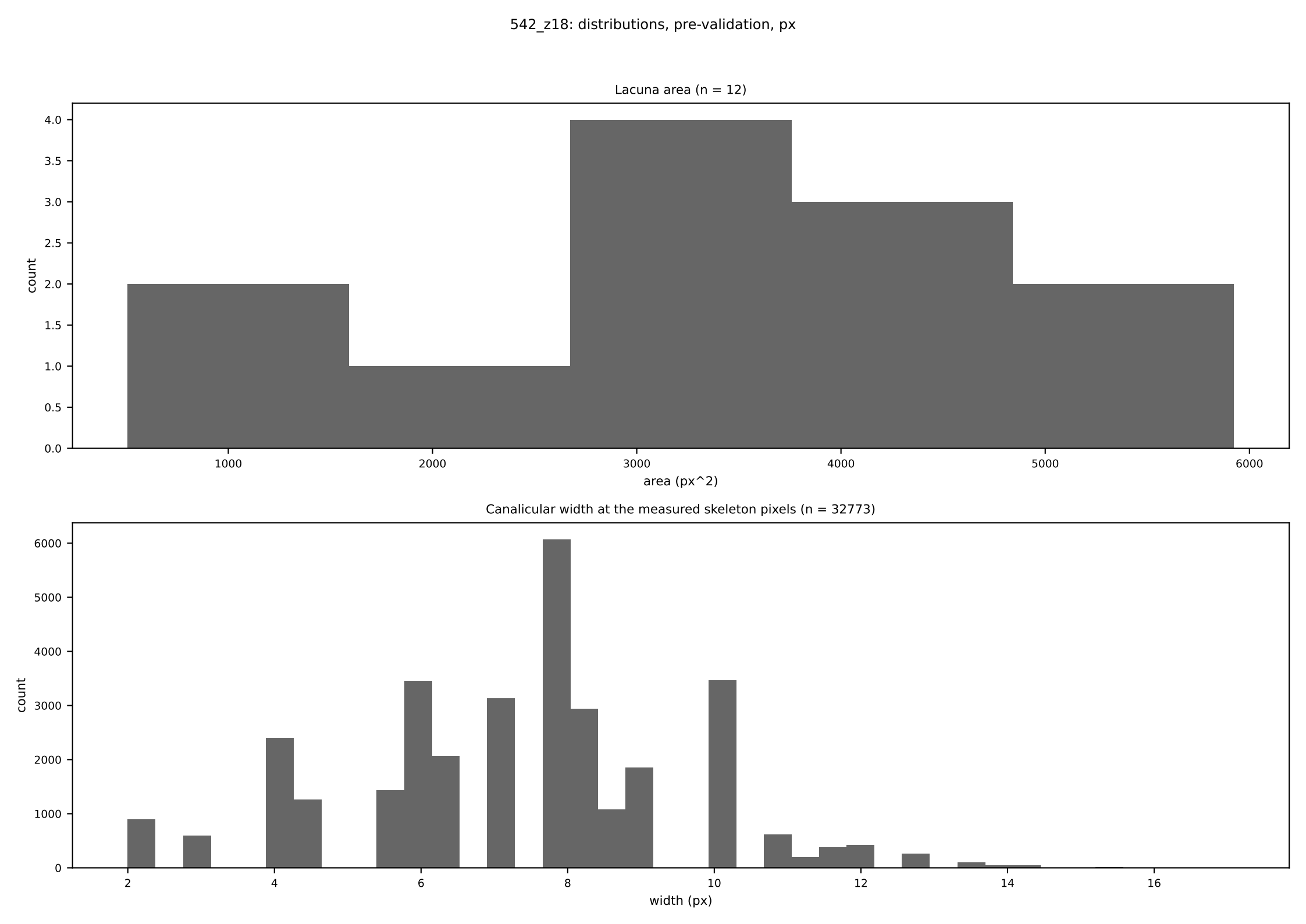
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter

1	142.811	4575	11641	0.673	0.6207	0.0424044	0.0450992	2.1007
2	625.725	16860	35430	0.9542	0.8861	0.0388493	0.0393452	2.8767
3	377.019	13152	31926	1	1	0.0234945	0.0266867	1.5914
4	337.706	12118	29514	1	1	0.0386202	0.0408281	2.3689
5	371.806	13199	31872	1	0.9956	0.032957	0.0345758	2.4206
6	308.593	11379	28385	1	1	0.026628	0.0291703	2.2684
7	382.877	12760	30245	1	1	0.0402038	0.046983	2.0894
8	306.492	10697	26160	1	1	0.0290736	0.0355122	2.6102
9	437.262	14880	35351	1	1	0.0233871	0.0240729	1.6009
10	326.836	11837	29271	1	1	0.035482	0.0427044	3.3656
11	263.48	9968	22850	0.9798	0.8783	0.0223716	0.0277899	1.8977
12	108.77	5218	13093	0.8837	0.75	0.0197394	0.0229894	3.6775

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	187	370	222.167	597.558	4	7	5
2	621	1151	741.198	1585.86	22	20	22
3	202	490	336.622	951.938	7	11	15
4	342	805	521.404	1386.08	10	16	19
5	322	865	484.505	1246.39	6	15	16
6	219	557	337.007	944.536	7	12	13
7	423	1257	583.902	1597.7	10	20	22
8	164	312	333.652	1032.28	8	12	15
9	312	550	398.777	971.051	12	9	13
10	348	917	464.848	1402.74	10	16	17
11	124	237	248.551	713.127	5	9	8
12	77	167	107.113	328.451	3	3	5


542_z18: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 32773)
count
width (px)